

433-HW2

Alex Caramanico

September 12, 2022

Problem 1

```
x = c(7,12,3,9,8) # test vector
# find arithmetic mean to compare to geometric and harmonic means
cat('The arithmetic mean is', mean(x), '\n')

## The arithmetic mean is 7.8

product = 1 # initialize product of vector values as 1
for (value in x) { # for loop to cycle through each value in vector
  # multiply each value by the product, the product updating each loop
  product = value * product
}
# apply transformations to the product to obtain geometric mean
cat('The geometric mean is', product^(1/length(x)), '\n')

## The geometric mean is 7.107987

# calculate harmonic mean with sum function and transformations
cat('The harmonic mean is', length(x)/sum(1/x))

## The harmonic mean is 6.284289
```

Problem 2

```
# sum of every 3rd element of a vector y
y = c(1,3,5,7,10,12,15) # test vector
sum_o_every_third = 0 # initialize sum as 0
for (i in 1:length(y)) { # for loop to traverse through each index in y
  if (3*i <= length(y)) { # ensure every third index is within the vector
    sum_o_every_third = y[3*i] + sum_o_every_third # add every third value to sum
  }
}
sum_o_every_third

## [1] 17
```

Problem 3

```
# for loop to find minimum of a vector z
z = c(2,5,2.2,7,12,1.9,16) # test vector
z_min = z[1] # initialize minimum value as the first value
for (i in 1:length(z)) { # create for loop which cycles through the indices of z
  if (z[i] < z_min) { # check if the value at the current index is less than the
    # current minimum
    z_min = z[i] # if condition is satisfied, update the minimum
  }
}
cat('The minimum value in the vector is', z_min)
```

```
## The minimum value in the vector is 1.9
```